

Emma Wiles (née van Inwegen)

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Academic Position & Affiliations

Isabel Anderson Career Development Professor, Boston University	2025-
Assistant Professor, Boston University	2024-
<i>Information Systems, Questrom School of Business</i>	
Visiting Scholar, Federal Reserve Bank of Minneapolis	Summer 2026
Digital Fellow, Initiative on the Digital Economy, Massachusetts Institute of Technology	2024-
Faculty Fellow, Digital Business Institute, Boston University	2025-

Education

PhD Management Science, Massachusetts Institute of Technology	2019-2024
Dissertation title: <i>Artificial Intelligence in Labor Market Matching</i>	
Committee members: <i>John Horton, Catherine Tucker, Dean Eckles</i>	
MS Management Research, Massachusetts Institute of Technology	2019-2022
BA Mathematics, Economics, University of Washington	2011-2015

Teaching Experience

BU IS223 Introduction to Information Systems	Fall 2024, 2025, 2026
MIT 15.818 MBA Level Pricing (TA)	Fall 2023
MIT 15.567 MBA Level Economics of Information: Strategy, Structure and Pricing (TA)	Spring 2023
MIT 15.575 PhD Level Economics of Information and Information Technology (TA)	Spring 2022

Professional Service

Platform Strategy Research Symposium, Scientific Committee	2026
Conference on Digital Experimentation, Technical Committee	2020, 2021, 2024, 2025 Questrom
Junior Faculty Brown Bag, Organizer	2026

Served as referee for:

Management Science, Information Systems Research, International Conference on Information Systems, MISQ, Quarterly Journal of Economics, Journal of Public Economics, Economics of Education Review, Journal of Human Resources, American Economic Journal: Applied Economics, American Economic Review: Insights

Fellowships, Honors, and Awards

Isabel Anderson Career Development Professorship (\$90,000)	2025-2028
Economics of AI Stripe Fellowship (\$10,000)	2025
PhD Thesis Award in Artificial Intelligence in Entrepreneurship and Management (AIEM) <i>University of Padova</i> (€5,000)	2024
Microsoft Research Grant for <i>AI and the Future of Work</i> (\$50,000)	2023
Zenon S. Zannetos Memorial Fellowship (MIT)	2020-2023
MIT Graduate Student Fellowship	2019-2020

Invited Seminars

McGill University, World Bank, Minneapolis Federal Reserve, University of Minnesota

Publications

[Algorithmic Writing Assistance on Jobseekers' Resumes Increases Hires](#) (with Zanele Munyikwa and John Horton), 2025

Published in Management Science

- Media coverage: MarketWatch, Business Insider, Yahoo News

There is a strong association between the quality of the writing in a resume for new labor market entrants and whether those entrants are ultimately hired. We show that this relationship is, at least partially, causal: a field experiment in an online labor market was conducted with nearly half a million jobseekers in which a treated group received algorithmic writing assistance. Treated jobseekers experienced an 8% increase in the probability of getting hired. Contrary to concerns that the assistance is taking away a valuable signal, we find no evidence that employers were less satisfied. We present a model in which better writing is not a signal of ability but helps employers ascertain ability, which rationalizes our findings.

[Why You Shouldn't Call AI Agents Employees](#) (with Megan Hsu, Lisa Kraye, Julie Bedard, and Matt Kropp), 2026

Published in Harvard Business Review

As organizations experiment with placing AI agents on org charts as “employees,” new research shows this framing has unintended consequences. In a large-scale experiment, anthropomorphizing AI reduced individual accountability, increased unnecessary escalation, lowered review quality, and heightened employee uncertainty about their roles—without improving adoption. The findings suggest the core challenge is not whether to deploy agentic AI, but how to redesign workflows, roles, and governance so humans remain clearly accountable while effectively supervising increasingly capable systems.

[Generative AI and the Temporary Upskilling of Knowledge Workers](#) (with Lisa Kraye, Mohamed Abadi, Urvi Awasthi, Ryan Kennedy, Pamela Mishkin, Francois Candelon, Daniel Sack), 2026.

Published in Nature Human Behaviour

“Upskilling” often refers to the process by which workers acquire and expand their skills, enabling them to perform different types of work as market demands change. This paper demonstrates that while generative artificial intelligence (GenAI) can act as an “exoskeleton,” enhancing workers’ capabilities while they attempt new skills, these gains are dependent on the continued use of the technology. When the “exoskeleton” is removed, little to no knowledge is retained independently, revealing that the newfound capabilities are temporary and reliant on the external support provided by GenAI. We run a randomized controlled trial on “upskilling” with GenAI by providing Boston Consulting Group (BCG) consultants with access and training in using ChatGPT to solve technical problems. We measure their performance on real data science tasks outside their skill sets, which cannot be independently solved by ChatGPT. Treated workers score 49, 20, and 18 percentage points higher than those in the control group on the three tasks and perform close to the level of real BCG data scientists on two of the three tasks. However, treated workers are no better at answering technical questions without the use of ChatGPT post-experiment, suggesting their demonstrated newfound technical capabilities do not imply knowledge acquisition.

[Minimum Wage Increases and Low-Wage Employment: Evidence from Seattle](#) (with Ekaterina Jardim, Mark Long, Robert Plotnick, Jacob Vigdor, Hilary Wething) 2022.

Published in American Economic Journal: Economic Policy

- Media coverage: The Economist, FiveThirtyEight, Los Angeles Times, New York Times, New York Times (The Upshot), Seattle Times, Washington Post

[Boundary Discontinuity Methods in the Presence of Policy Spillovers](#) (with Ekaterina Jardim, Mark Long, Robert Plotnick, Jacob Vigdor) 2022.

Published in the Journal of Public Economics

Working papers

[More, but Worse: The Marketplace Effects of AI-Generated Job Posts](#) (with John Horton), 2025.
Revise and Resubmit at Management Science

- Media coverage: New York Times

Generative AI makes it dramatically cheaper to produce the information that labor markets rely on for hiring—but cheaper communication need not improve matching. We consider the effects of the introduction of a generative AI tool in a large scale field experiment on an online labor market, which lowered employers' search costs by randomly offering them AI-written first drafts of their job post. The assistance was widely accepted and treated employers were 19 % more likely to post a job. It also saved employers time—treated employers spent 44% less time writing their job posts. Despite the substantial increase in job posts, there was no discernible increase in matches in the treatment group. We provide evidence that the treated job posts were more generic and less informative to jobseekers. This combination of increased job post volume and reduced informativeness diluted signals of employer seriousness. A time-cost accounting implies jobseeker application time costs are roughly six times larger than employer time savings per treated employer. In a subsequent platform roll-out expanding the access of the AI tool, posting responses attenuated while lower hiring conditional on posting remained, suggesting that learning and increased adoption did not eliminate these frictions.

[Putting AI on the Org Chart: Evidence on Delegation and Oversight](#) (with Megan Hsu, Julie Bedard, and Matt Kropp).

Motivated by the potential for large productivity gains from AI, firms are increasingly deploying agentic AI systems capable of independent action. Some firms have also begun formally integrating these agents into their organizational structures—assigning them designated roles and responsibilities, and in some cases explicitly referring to them as employees. This creates new challenges for manager's decisions about when to delegate and how to monitor work. In a survey of 1,261 managers we find that 23% already work in organizations where AI agents have been formally institutionalized on organizational charts. In a randomized experiment we provide those managers with identical documents containing built in errors, where we vary whether the drafts are presented as produced by an AI tool, an AI employee, or a human employee. Average effects to error catching are small. However, in the subgroup of managers whose organizations already have 'AI employees', presenting identical drafts as produced by an AI employee (versus an AI tool) reduces managers' accuracy by 17%, increases their reliance on additional review from others by 22 percentage points, and shifts their perceived accountability away from themselves and toward the AI system. The human employee condition shows that this is not simply a response to delegation, managers direct review increases for human employees and they are substantially less likely to seek additional review. These results suggest that formalizing AI as an organizational actor changes the allocation of oversight, with managers doing less direct verification and relying more on additional review. How organizations position AI should therefore be understood as a governance decision, rather than as a mere labeling choice.

[The Diminishing Returns to Human Recruiting in Online Labor Markets](#) (with John Fallon and John Horton)

Employers often rely on outside recruiters to find workers, but it is unclear whether human intermediaries add value when employers already have access to algorithmic screening tools. We study a randomized experiment in a large online labor market that assigned human recruiting assistance to 83,017 vacancies. The recruiters increased the search and screening intensity of treated job posts: treated job posts had more recruiting and more applicant screening done on their behalf, resulting in 5% more applicants. Employers of treated job posts redirected their own screening efforts toward recruiter-sourced candidates and away from applicants who found the post on their own. Despite the increase in search intensity, treated employers were no more likely to hire than employers of control posts with access only to algorithmic tools. Match quality appears at best, no better for the hires that resulted from the treated job posts, and treated employers were less likely to return to the platform to post a second vacancy after the experiment. We develop a model of delegated recruiting in which recruiters and employers rely on a common noisy signal; it predicts that recruiting adds value when the recruiter brings information the algorithm lacks and can subtract value when the two parties' assessments are correlated. Consistent with the model's screening mechanism, recruited applicants are positively selected on observable

characteristics yet do not improve hiring outcomes. These findings suggest that as algorithmic screening improves, the scope for intermediaries to add value shrinks because it becomes harder to access independent information.

Research in Progress

Workers Responses to Price Uncompetitiveness: Evidence from a Field Experiment (with Apostolos Filippas and John Horton)

If and how to regulate online marketplaces is an open question important to both platform designers and policy makers. Using a large field experiment in an online labor market, we analyze the effects of a platform minimum wage. Workers were randomly assigned individual price floors which prevented treated workers from bidding hourly rates below their floor. Workers for whom the floor was likely binding—those historically bidding below the floor—suffered a decline in job-finding probability(30%), but higher wages conditional upon being hired(9%). Treated workers made lower earnings overall, but higher earnings conditional on working at least one hour on the platform. Despite a job being “worth more” if hired, affected workers lowered their search intensity. They did not move to the “uncovered sector”—jobs with a fixed price rather than an hourly wage, nor did they direct their search to better fitting jobs. They were also more likely to exit the platform. After the conclusion of the experiment, the platform rolled out the \$3 per hour minimum wage platform wide, allowing us to observe the the employment outcomes and job search behavior in equilibrium.

AI Competence as Human Capital (with Katelyn Cranney)

Generative AI can raise productivity, but realized gains depend on how workers use the tool. We study AI competence as a form of human capital defined by the practical ability to organize work with a model, evaluate its output, and retain judgment while using it to extend one’s existing skills. In a preregistered lab-in-the-field experiment, 332 full-time management consultants whose jobs did not routinely require coding were assigned a difficult Python data-analysis task with or without access to a Gen AI tool. AI access increased scores by 34 percentage points, raised completion by 7 percentage points, reduced time on the task by 12%, and improved debugging efficiency. We open the black box of these gains by combining Gen AI transcripts, code-execution logs, task outputs, and surveys. Using an ethnography-inspired agentic coding procedure, we identify nine distinct AI-collaboration practices that capture how workers frame requests, decompose problems, rely on generated code, edit independently, verify outputs, and manage scope. These practices explain substantial heterogeneity in overall performance. Gains concentrate when workers engage proactively with AI while retaining substantive judgment. Finally, we show that productivity gains and workers’ interpretation of those gains are distinct. AI access does not significantly increase average confidence, trust, enjoyment, or behavioral trust in AI, and belief responses vary substantially by gender and prior coding experience. The results suggest that firms should treat AI training as human-capital investment, teaching workers not only how to prompt, but how to divide labor with AI, evaluate its output, and build confidence in using it well.

Dimensions of Adoption: The Integration and Governance of AI Agents in U.S. Firms (with Innessa Colaiacovo)

We provide some of the first systematic evidence about the extent of agentic AI integration and governance at U.S. firms based on a survey of 1,500 full-time senior managers. We find that just over half of all managers report that their organization has at least one AI agent built into work processes, but fewer than half of these managers report their organization has a formal policy to govern their use or monitor their outputs. We then show that agent-adopting organizations vary in the intensity of both their integration and governance of AI agents, and intensive integration of agents is not always accompanied by enacting policies to govern them (and vice versa). These results highlight that agentic AI is a real phenomenon in U.S. workplaces, but that classifying all firms where agents are present as “adopters” may conflate firms that are and are not making co-occurring adjustments to their workforce and AI governance practices.

Invited presentations

Columbia’s Management, Analytics, and Data Conference	2024
National Association for Business Economics Tech Economic Conference Lightning Talk	2023
Wharton, Conference on Business & AI Presentation	2023

NBER Summer Institute Digital Economics and AI Lighting Round Talk	2023
Conference on the Economics of Information & Communication Technologies Presentation	2023
ASSA/AEA, Labor and Employment Relations Association Presentation	2023
NBER Digitization Tutorial	2022
NBER Economics of Privacy Conference	2022
INFORMS, Platforms Presentation	2022
Workshop on Information Systems (WISE) & Economics, Platforms Presentation	2022
Conference on Digital Experimentation, Presentation	2020

Personal Details

Language: English (Native)
Citizenship: USA